

## Practical Question for Practice [26-05-2026]

Topic: **Boolean Filtering & Indexing in DataFrame - Python**

Class XII – Informatics Practices (065)

Consider the following DataFrame 'df1':

|   | ProductID | ProductName     | Category    | Brand    | Price | Quantity | Rating |
|---|-----------|-----------------|-------------|----------|-------|----------|--------|
| 0 | 101       | iPhone 15       | Mobile      | Apple    | 799   | 15       | 4.8    |
| 1 | 102       | Galaxy S23      | Mobile      | Samsung  | 749   | 10       | 4.5    |
| 2 | 103       | MacBook Air     | Laptop      | Apple    | 999   | 8        | 4.9    |
| 3 | 104       | Dell XPS        | Laptop      | Dell     | 1200  | 5        | 4.2    |
| 4 | 105       | AirPods         | Audio       | Apple    | 249   | 50       | 4.7    |
| 5 | 106       | Sony WH-1000XM5 | Audio       | Sony     | 349   | 12       | 4.8    |
| 6 | 107       | iPad Pro        | Tablet      | Apple    | 899   | 0        | 4.6    |
| 7 | 108       | Kindle          | E-Reader    | Amazon   | 139   | 20       | 4.4    |
| 8 | 109       | Apple Watch     | Wearable    | Apple    | 399   | 18       | 4.5    |
| 9 | 110       | Gaming Mouse    | Accessories | Logitech | 79    | 25       | 4.1    |

### Questions:

1. Display the first 5 records of the inventory.
2. Display all products that belong to the '**Mobile**' category.
3. Show all products where the **Price** is greater than **800**.
4. Display only the ProductName and Brand for items where **Quantity** is 0 (Out of stock).
5. Find all '**Apple**' products that are priced below **500**.
6. Display products that are either in the '**Laptop**' category or the '**Tablet**' category.
7. Show all products where the **Rating** is above **4.5** but the **Price** is less than **900**.
8. Display products that are NOT from the brand '**Apple**'.
9. Find all products whose **ProductName** contains the word '**Watch**'.
10. Display products that have a **Price** higher than the **average price** of the entire inventory.
11. Show products whose **ProductName** starts with the letter '**S**'.
12. Filter products where the Price is one of the following: **[249, 349, 799]**.  
(Hint: Use .isin())
13. Increase the **Quantity** by **10** for all products in the '**Accessories**' category.
14. Create a new column called '**InventoryValue**' which is Price \* Quantity.
15. For any product with a **Rating** below **4.3**, change the ProductName to include the prefix '**Refurbished -** '.
16. Update the **Price** of all '**Laptop**' items by applying a **10% discount**.
17. Identify the ProductName of the item that has the **highest Rating** in the entire store.
18. Display a filtered DataFrame showing products where the **Stock Value** (Price \* Quantity) is greater than **10,000**.
19. Find products where the **Brand** is '**Apple**' and they are **either** in 'Mobile' **or** 'Wearable' categories.
20. Create a column '**Status**':
  - a. If Quantity > 20 → '**High Stock**'
  - b. If Quantity between 1 and 20 → '**Available**'
  - c. If Quantity == 0 → '**Out of Stock**'

\*\*\* END OF PAPER \*\*\*